COMPENDIUM OF INDIAN STANDARDS ON

PERSONAL PROTECTIVE EQUIPMENT (PPE)

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NEW DELHI

FOREWORD

Handbook on Personal Protective Equipment (PPE)

Personal Protective Equipment (PPE) plays a crucial role in ensuring safety across various industries, safeguarding workers from potential hazards. With the growing emphasis on workplace safety, the demand for reliable and standardized PPE has significantly increased. It is essential to establish uniform quality requirements and testing criteria for assessing PPE to ensure its effectiveness, durability, and compliance with safety regulations.

The **Bureau of Indian Standards (BIS)**, through its **Chemical Division Council (CHDC)**, is actively involved in the preparation of Indian Standards in the field of Personal Protective Equipment (PPE). The technical committee responsible for the standardization of PPE is the **Occupational Safety and Health Sectional Committee, CHD 08**.

This handbook aims to provide technical guidance on published Indian Standards related to PPE, helping stakeholders access relevant standards for reference.

HEAD PROTECTIVE EQUIPMENT

Safety helmets are an essential component of personal protective equipment (PPE), designed to protect individuals from head injuries in various hazardous conditions. India has established several standards to ensure the quality and reliability of helmets used for different purposes. Below is a detailed discussion of the key standards for head protective equipment

List of Indian Standards on Head Protective Equipment:

- **IS 11207 : 1983** — *Specification for Helmets for Mountaineers (Reaffirmed in 2021)*

IS 11207:1983, Specification for Helmets for Mountaineers, establishes safety and performance benchmarks for helmets intended to protect climbers from head injuries caused by falling objects or impacts during falls. This Indian Standard ensures that mountaineering helmets meet essential criteria for material quality, construction, and rigorous testing to provide reliable protection in hazardous environments.

Key tests include shock absorption, where a wooden striker dropped from 2.5 metres must not transmit over 20 kN of force, and a penetration test with a pointed striker to prevent contact with the headform. The retention system is tested for strength and stretch, while additional tests evaluate rigidity, flammability, and corrosion resistance, ensuring helmets perform effectively under varied environmental conditions.

- **IS 2745 : 1983** — *Non-Metal Helmet for Firemen and Civil Defence Personnel (Reaffirmed in 2020)*

In high-risk jobs like firefighting and civil defence, **helmets play a crucial role in safety**, protecting against impacts, heat, and other hazards. A good helmet should offer **solid impact protection, be comfortable to wear, and resist fire and electrical risks**, keeping the wearer safe in challenging conditions.

IS 2745 mandates various tests to confirm the helmet's safety. These include shock absorption and penetration resistance tests to assess durability in high-impact situations, a water absorption test to ensure low water intake, and a flammability test to confirm fire resistance. Electrical resistance tests also verify the helmet's insulation against electrical hazards.

- **IS 9562 : 1980** — *Specification for Non-Metal Helmet for Police Force (Reaffirmed in 2022)*

The document IS 9562: 1980 specifies requirements for **non-metal helmets** designed for **police use**. It details material, construction, and testing requirements to ensure effective head protection for police in situations such as **riot control**. This non-metal helmet is built to withstand impacts from blunt objects and other hazards encountered during active duty. It includes a **hard outer shell** made of impact-resistant, non-metallic materials, **protective padding of expanded polystyrene, and a retention system** with a headband, chin strap, and cradle for secure fitting.

This standard mandates performance tests:

- Shock Absorption – Maximum transmitted force must not exceed 20 kN under various temperature and durations.
- Penetration Resistance – Two methods used are plate load test & plumb bob
- Water Absorption – Helmets should absorb less than 5 percent of their mass in water.
- Other tests, which are included are Strength of Retention System, Rigidity Test, Firing Test, Flammability Resistance.

· **IS 2925 : 1984** — Industrial Safety Helmets ((Reaffirmed in 2022)

IS 2925: 1984 outlines the specifications for **industrial safety** helmets that are essential for protecting workers against head injuries in environments prone to accidents, such as construction sites, mines, and industrial facilities. The standard covers helmets designed to shield the skull from impacts due to falling objects and other similar hazards.

Consumers expect these helmets to offer robust **impact resistance**, durability, and comfort. Key quality expectations include a snug fit to prevent slippage, lightweight design for ease of wear, and sufficient ventilation to keep the wearer cool under various environmental conditions. It mandates the use of non-metallic, penetration-resistant materials for the shell and details requirements for the harness system inside the helmet, including headbands and cradle. To enhance user comfort and safety, it incorporates features such as adjustable straps and sweat-resistant padding. Additionally, the standard allows for optional ventilation holes that do not compromise the structural integrity of the helmet. The safety helmets are rigorously tested for flammability resistance, ensuring they do not exacerbate potential fire hazards.

· **IS 4151 : 2015** — Helmets for Motorcycle Riders

Protective helmets for motorcycle riders are essential **personal protective equipment**, designed to reduce the risk of **head injury** during **road accidents**. Since this product is intended for everyday use by two-wheeler riders, these helmets have to necessarily provide critical protection against **impact and abrasion hazards**.

The standard mandates **impact absorption** tests with impact points and use of a kerb stone anvil, increased impact velocity and head injury criteria, to ensure the helmet effectively reduces head injury risks in various scenarios. The standard ensures safety and comfort by mandating impact absorption tests with a kerb stone anvil, secure chin straps with micro-slip and abrasion tests, and comfort padding made from sweat-resistant, non-irritant materials. It also allows ventilation and includes side impact tests under varied temperatures and humidity conditions.

EYES AND EARS PROTECTIVE EQUIPMENT

Protection for the eyes, ears, and face is essential across various occupational environments to ensure worker safety and prevent injuries or long-term health issues.

List of Indian Standards on Eyes and Ears Protective Equipment

- **IS 8521 (Part 1) : 2022** — *Eye and Face Protection for Occupational Use*

The product defined in IS 8521: 2022 is eye and face protection for occupational use, covering safety spectacles, goggles, and face shields used in workplaces to protect against common hazards like impacts, optical radiation, dust, splashing liquids, molten metals, heat, flame, harmful gases, and vapours. These protective gear ensure the safety of workers' eyes and faces in environments such as manufacturing, welding, and chemical handling (IS 8521). The standard addresses these expectations through specific requirements and testing methods:

- Geometrical optical properties ensure clear vision by setting tolerances for spherical and cylindrical power.
- Physical and mechanical properties like impact resistance are tested to guarantee durability under hazardous conditions.
- Temperature performance is validated between −5°C to +55°C to ensure the protector's effectiveness in extreme environments.

- **IS 9167 : 1979** — *Specification for Ear Protectors*

Ear protectors are devices designed to prevent hearing damage by reducing exposure to high noise levels, particularly in environments such as industrial settings. The primary types include ear-plugs, ear-muffs, and helmet-integrated ear protectors. Each type offers either broad or frequency-specific noise reduction, tailored to the sound environment, ensuring effective and comfortable protection across a range of applications.

The core test for these devices is the Sound Attenuation Test, which measures the effectiveness of noise reduction across specific frequency ranges. To validate the effectiveness and resilience of ear protectors, the standard includes a series of rigorous performance tests that assess various quality attributes. These tests include:

- Damp Heat and Temperature Change Tests: Ensure that ear protectors maintain their structure and functionality despite temperature fluctuations and humidity exposure.
- Low-Temperature Impact Test: Confirms that ear protectors are resistant to damage from impacts at low temperatures.
- Cleanability Test: Ensures the material can be cleaned effectively without degrading, maintaining hygiene and safety.
- Salt Mist Test: Evaluates resistance to corrosive elements, particularly important in environments with salt or other corrosive substances.
- Headband Extension Test: Verifies the headband's durability and elasticity under repeated stretching.

RESPIRATORY PROTECTIVE DEVICES

Face mask protective equipment includes surgical masks, respirators (e.g., N95, FFP2), cloth masks, and PAPRs, designed to protect against airborne particles and pathogens. Standards ensure filtration efficiency, breathability, and safety for medical and industrial use.

<u>List of Indian Standards on Face Mask Protective Equipment:</u>

- **IS 9473 : 2002** — *Respiratory protective devices — Filtering half masks to protect against particles*

The product defined in IS 9473:2002 is respiratory protective devices specifically filtering half masks designed to protect the wearer from particles such as solid aerosols, liquid aerosols, or water-based aerosols. These masks are commonly used in environments with dust, fumes, or other airborne particulate matter, particularly in occupational settings like construction, mining, or chemical industries.

The standard outlines testing methods for filtration efficiency, breathing resistance, and total inward leakage to ensure masks meet the expected performance and safety criteria. For instance, sodium chloride and paraffin oil tests assess the mask's ability to filter solid and liquid aerosols. Practical performance tests ensure the mask's usability under real-world conditions, while material specifications ensure that the mask is both durable and skin safe.

- **IS 14166 : 2024** — *Respiratory protective devices full — Face masks — Specification*

This Indian Standard, IS 14166: 2024, specifies requirements for **full face masks used as part of respiratory protective devices**, excluding escape apparatus and diving apparatus. The need for this standard arises from the critical role these masks play in **occupational safety and health**, protecting wearers from hazardous ambient atmospheres by covering the eyes, nose, mouth, and chin and providing an adequate seal. The standard ensures that this seal remains effective even with movement and speech, and on both dry and moist skin.

The standard mandates comprehensive tests for full face masks, including leak tightness, flammability, temperature resistance, inward leakage, and CO_2 content. It also tests harness and connector strength, impact resistance of visors, valve performance, breathing resistance, and thermal radiation resistance, ensuring safety, durability, and comfort under realistic and extreme conditions.

- **IS 14746 : 2025** — *Respiratory protective devices — Half masks and quarter masks — Specification*

These respiratory protective masks are meant to cover the nose and mouth (half masks) or part of the face below the nose (quarter masks), offering filtration and protection against airborne contaminants. They are designed to protect against particulates, certain gases, and vapors, depending on the intended application and filter type.
Half masks and quarter masks are commonly used in industries such as construction, mining, and chemical processing, where respiratory protection is needed.

The standard outlines specifications for the materials, construction, performance, and testing of half masks and quarter masks designed to protect users from respiratory hazards. Masks must meet specific criteria for filtration efficiency, breathability, and durability under defined test conditions. It also encompasses design elements such as face seal, fit, comfort, and stability. Proper sealing is crucial to ensure that the mask effectively prevents contaminated air from bypassing the filter.

- **IS 15803 : 2008** — *Respiratory protective devices — Self-contained closed circuit breathing apparatus chemical oxygen (KO2) type, self-generating, self-rescuers — Specification*

This Indian Standard, IS 15803: 2008, addresses a crucial piece of life-saving equipment: self-contained closed-circuit breathing apparatus of the chemical oxygen (K02) type, specifically self-rescuers. The need for such a standard arises from the fact that breathing apparatus is paramount for survival in irrespirable atmospheres. These K02 type self-rescuers are increasingly replacing filter self-rescuers and are designed for escape by individuals, particularly miners, from hazardous environments following events like explosions, allowing them to reach safety within a reasonable timeframe.

The standard for self-rescuers includes tests using breathing machines to assess gas composition, breathing resistance, and surface temperature. It mandates human performance trials, extreme temperature exposure, leak tightness checks, flammability tests, and mechanical strength assessments, ensuring reliability, safety, and functionality under both laboratory and real-world emergency conditions.

- **IS 10245 (Part 1) : 1996** — *Breathing Apparatus: Part 1 Closed Circuit Breathing Apparatus (Compressed Oxygen Cylinder) — Specification (First Revision)*

This Indian Standard, IS 10245 (Part 1) : 1996, specifies the requirements for closed-circuit breathing apparatus utilising compressed oxygen cylinders. The need for such breathing apparatus arises from the necessity for individuals to operate safely in environments where the atmosphere is irrespirable or contains poisonous substances. These situations can occur for short or extended periods, demanding that the wearer retains full physical and mental capacity.

The apparatus is crucial as rescue equipment, anti-gas devices, respiratory protection in smoke and gaseous environments, and as gas masks in various industries such as mines, gasworks, chemical factories, firework production, shipyards, fertilizer plants, metallurgical operations, oil refineries, and oil tankers. Furthermore, it is essential for fire brigades, municipalities, defence personnel, and mountaineers.

The standard ensures breathing apparatus safety through leak tightness tests, valve performance checks, and material durability assessments. It includes practical tests simulating various work conditions, evaluates breathing resistance and CO_2 management using lung simulators, and tests facepiece fit, canister efficiency, and climatic performance, ensuring reliability across diverse operational scenarios.

- **IS 10245 (Part 2) : 2023** — *Self-Contained Open Circuit Breathing Apparatus(Second Revision)*

A breathing apparatus is one of the most important lifesaving equipment and the self-contained open circuit breathing apparatus is widely used in work situation where the risk of over pressurization of the cylinders due to hot environmental conditions is low and this standard specifies its minimum performance requirements.

The performance of this apparatus is ensured through its Designs, Components/Materials, Procedures opted for Cleaning and Disinfection, Mass, Water Immersion during submerged Condition, Work Ability of demountable Connections, Couplings, Strength of Connections, Capability of standing with drop for the Body Harness and the Practical performance. Further various parameters related to Resistance for Temperature and Flammability, Pressure Reducer, Warning Devices, Leak Tightness, Water Immersion and Resistance to collapse of Breathing Hose/Breathing Resistance are ensured so that the equipment performs as per desired conditions and its purpose.

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- **IS 10245 (Part 3) : 1999** — *Fresh Air Hose and Compressed Air Line Breathing Apparatus — Specification (First Revision)*

Breathing apparatus enables a person to remain in irrespirable and poisonous atmosphere for long or short periods and to still retain his full physical and mental capacity. It is also known as rescue apparatus anti-gas apparatus, respirator, smoke helmet and gas mask. The apparatus is required in mines, gas works, chemical factories, iron works, steel plants and oil refineries etc. It may also be used by fire brigade, municipality, army, navy and air force personnel and mountaineers.

The requirements specified in the Standard ensures that materials used in the construction has adequate mechanical strength, durability, resistance to corrosion and resistance to deterioration by heat, fire and water. It is also taken care that the apparatus shall be such that it may be worn without avoidable discomfort with hood completely covering the head, neck and portions of the shoulders and prevent the leakage. The Standard also ensures that the apparatus shall be provide with a harness or belt to prevent a pull on the breathing tube or on the mouthpiece. Importantly this Standard also put a check on carbon dioxide content of the inhaled air.

- **IS 10245 (Part 4) : 1982** — *Escape Breathing Apparatus (Short Duration Self-Contained Type)*

This Indian Standard, addresses the critical need for escape breathing apparatus of the short duration, self-contained type. Such apparatus is essential for enabling individuals to safely evacuate areas contaminated by dangerous dusts, gases, fumes, or vapours, which can be found in various industrial settings like mines, gas-works, chemical factories, iron works, steel plants, oil refineries, and even in operations by fire brigades, municipalities, and armed forces.

The standard mandates tests for open- and closed-circuit escape breathing apparatus, including inward leakage on diverse users, simulated breathing performance, and canister durability. Practical tests assess comfort, mobility, and functionality. Additional evaluations cover relief valve leakage, system leak tightness, water resistance, and inspiratory and expiratory breathing resistance.

BODY PROTECTIVE EQUIPMENT

Body protective equipment is vital for safeguarding workers against various occupational hazards, such as exposure to heat, flame, chemical agents, and physical injuries.

List of Indian Standards on body protective equipment:

- **IS 15748 : 2022** — *Clothing to Protect Against Heat and Flame — Minimum Performance Requirements (First Revision)*

Protective clothing is essential for workers exposed to heat and flame hazards in industrial settings. This is designed to safeguard the wearer's body, except hands, against heat, flames, and molten metal splashes. Key quality parameters include flame retardancy, high tensile strength, resistance to molten metal splashes, and breathability to allow moisture wicking.

This standard covers the general and the design requirements and additional design requirements for molten splash protective garments. The method of sampling, Pre-treatment, aging and conditioning temperature and humidity. General performance requirements include Heat resistance at a temperature of (180 ± 5) °C and (260 ± 5) °C, flame spread etc. the various physical requirements include Tensile strength, tear, burst and seam strength and also Heat transmission performance requirements are specified.

- **IS 16890 : 2024** — *Firefighter Suits*

This Indian Standard, IS 16890:2024, specifies requirements for protective clothing designed for firefighters, acknowledging the extensive and life-threatening role firefighters play in rescuing lives and saving properties in hazardous environments. The standard recognises the intensive physical work demanded by this occupation and the necessity of providing suitable personal protective equipment (PPE), of which the firefighting suit is a critical component.

The standard outlines several important testing requirements to ensure the protective capabilities of the firefighter clothing. Flame resistance is a crucial aspect, with tests for flame spread on the outer garment and its lining, as well as on seams and wristlet materials. These tests assess factors like flaming to edges, hole formation, molten debris, after-flame time, and afterglow time. The standard also specifies requirements for heat transfer under flame and radiant exposure, with different performance levels for Category 1 and Category 2 clothing. Tests for contact heat resistance, residual strength of material after radiant heat exposure, and overall heat resistance of materials and hardware are also mandated. Mechanical performance is evaluated through tensile strength and tear strength tests, with higher requirements for Category. Furthermore, the standard includes tests for cleaning-shrinkage resistance, liquid-chemical penetration resistance, water-penetration resistance, and water-vapour resistance. These comprehensive testing requirements ensure that the protective clothing can withstand the demanding conditions of firefighting.

- **IS 16874 : 2018** — *Firefighter Gloves*

This Indian Standard addresses the critical need for high-quality protective gloves for firefighters. It has been observed that gloves are often a misunderstood item within firefighters' clothing ensembles. However, hand protection during firefighting is of paramount importance because burns and other hand injuries still constitute a significant portion of all fire ground injuries.

The standard for firefighters' gloves specifies rigorous tests to ensure protection against thermal, mechanical, chemical, and ergonomic hazards. Thermal tests assess resistance to flame, radiant, and conductive heat. Mechanical tests include abrasion, cut, tear, puncture, and seam strength. Barrier tests ensure resistance to water and harmful chemicals. Ergonomic tests evaluate dexterity, grip, and ease of donning. These requirements ensure gloves provide reliable protection, durability, and functionality without compromising the wearer's ability to perform critical firefighting tasks.

- **IS 15809 : 2017** — *High-Visibility Warning Clothes*

High visibility warning clothes, often called "hi-vis" or "hi-viz" clothing, are designed to make the wearer highly visible in low-light or high-risk environments, improving safety by helping them stand out. This standard specifies photometric and physical performance requirements for retroreflective and combined performance materials. Commonly used in industries like construction, road work, warehousing, and emergency services, these garments often feature bright colors and reflective strips that enhance visibility during the day and night. This standard specifies requirement for three classes of High visibility warning clothes based on level of conspicuity (easy to see).
§ Bright, Fluorescent Colors: Fluorescent colors make the clothing stand out in daylight, twilight, and even in shadowed areas.
§ Reflective Strips: These strips are often positioned around the torso, arms, and legs for maximum visibility when illuminated by headlights or other light sources.
§ Comfort and Durability: Hi-vis clothing is usually made from materials that are breathable, flexible, and durable for outdoor work.

- **IS 18849: 2024 —** *Work wear for Cement Workers*

This Indian Standard addresses the critical need for protective and comfortable work wear specifically designed for cement workers. The cement industry presents a challenging work environment with various hazards, most notably exposure to dust. Therefore, this standard is important as it specifies the minimum requirements for the protective clothing intended to be worn during work with cement and associated activities.

The standard specifies stringent testing requirements for workwear materials to ensure durability and protection. The outer shell must be polyester with defined weave, thread count, mass per unit area, and Taber abrasion resistance. It must meet minimum ratings for colour fastness, dimensional stability, and pilling resistance. The inner mesh layer is tested for mass and bursting strength. The combined multilayer fabric must pass tearing, tensile, air and water vapour permeability, hydrostatic pressure, and dust resistance tests. Seam strength must exceed 200 N.

- **IS 16655 : 2017** — *Protective Clothing for Welding and Allied Processes*

Protective clothing is to shield wearer from potential dangers like burns, sparks, spatter, electric shock, and radiation. This standard aims to ensure the safety and protection of workers from various hazards associated with welding operations, including short-term exposure to heat, sparks, molten metal, and heat transfer by radiation.

The standard outlines essential criteria for the materials used in protective clothing, emphasizing their durability, flame resistance, and thermal insulation properties. It specifies the types of fabrics suitable for welding protective gear, which must be capable of withstanding temperatures and preventing burns or injuries. Additionally, the standard addresses the design and construction of the clothing, ensuring that it provides adequate coverage and protection without compromising the wearer's mobility and comfort. IS 16655:2017 also includes guidelines for testing methods to evaluate the performance of the protective clothing, focusing on aspects such as heat resistance, flame spread, and the ability to withstand mechanical stresses.

- **Personal Fall Arrest System — IS 3521**

The Indian Standard IS 3521 lays down specifications for Personal Fall Arrest Systems (PFAS) to ensure worker safety in height-related tasks. The standard covers the design, testing, and performance requirements of fall protection equipment, including harnesses, lanyards, energy absorbers, connectors, and anchorages.

IS 3521 (Part 1) : 2021 'Full Body Harness (Fourth Revision)' — Full body harnesses are required to protect persons from injury by arresting the fall in the event of the user losing his/her balance or support and falling from the place of working either at an elevation or in closed locations or containers.

IS 3521 (Part 2) : 2021 'Lanyards and Energy Absorbers' — Lanyards and energy absorbers are used together as a connecting subsystem in personal fall-arrest systems (PFAS).

IS 3521 (Part 3) : 2021 '*Self-retracting Lifelines*' — Self-retracting lifelines are used as a connecting sub-system in personal fall-arrest systems (PFAS).

IS 3521 (Part 4) : 2021 '*Vertical Rails and Vertical Lifelines Incorporating a Sliding-type Fall Arrester*' — Vertical rails and vertical lifelines incorporate a sliding-type fall arrester and constitute a personal fall-arrest system (PFAS) when connected to a full-body harness.

IS 3521 (Part 5) : 2021 '*Connectors with Self-closing and Self-locking Gates*' — Connectors are used in personal fall-arrest systems (PFAS), such that, if an arrest takes place, the arresting force will not exceed 6 kN.

IS 3521 (Part 7) : 2021 '*Single Point Anchor Devices*' — This standard specifies requirements, test methods, and marking, labelling and packaging, as appropriate, of both permanent and temporary single-point anchor devices exclusively for the attachment of personal protective equipment (PPE) for protection against falls from a height for fall arrest, work positioning and travel restriction (work restraint).

IS 3521 (Part 8) : 2021 '*Flexible Horizontal Lifeline Systems*' — This Standard specifies design and performance requirements, test methods, user instructions, marking and labelling as appropriate, of flexible horizontal lifeline systems for use at any one time by up to three persons, exclusively for the attachment of personal protective equipment for protection against falls from a height. It does not stipulate designs for flexible horizontal lifelines, except for design limitations that are necessary for safe and durable service.

IS 3521 (Part 9) : 2021 '*Descending Devices*' — Descending devices are intended to be used in conjunction with an appropriate descent line and other equipment, for example, an appropriate harness or rescue loop and a reliable anchor, to enable a person to descend from one position to another, either on their own or assisted by another person.

FOOT AND LEGS PROTECTIVE EQUIPMENT’S

Foot and leg protection is critical for workers in industries where they face risks such as heavy objects, sharp materials, chemicals, or extreme temperatures. Below is an overview of these standards and their applications.
List of Indian Standards on Wear Safety Shoes Protective equipment:

- **IS 15298 (Part 2) : 2024** — *Safety Footwear*

This Indian Standard is crucial as it specifies the basic and additional requirements for safety footwear used for general purposes. The need for such a standard arises from the necessity to protect wearers from injuries that could occur due to accidents, encompassing mechanical, slip, thermal risks, and ergonomic behaviour. The importance of this standard is underscored by the fact that slips, trips, and falls are major causes of workplace accidents, leading to personal and financial costs.

- **IS 15298 (Part 3) : 2024** — *Protective Footwear*

This Indian Standard specifies the basic and additional requirements for protective footwear designed for general purpose use. The need for such a standard arises from the necessity to protect individuals from various workplace hazards, including mechanical risks, slip resistance, thermal risks, and ergonomic factors. It also extends to cover customised protective footwear, acknowledging the need for individually adapted solutions.

- **IS 15298 (Part 4) : 2024** — *Occupational Footwear*

The IS 15298 (Part 4) : 2024 standard on occupational footwear addresses a critical need for ensuring the safety and well-being of workers across various industries. Occupational footwear serves as essential personal protective equipment (PPE), designed to shield wearers from potential workplace hazards that could lead to injuries. These hazards encompass mechanical risks, the danger of slips, thermal risks, and ergonomic considerations, underscoring the importance of having reliable and effective footwear.

- **IS 1989 (Part 1) : 1986** — *Leather Safety Boots for Miners*

When choosing safety boots for mining, factors like protection, durability, and comfort are essential. Miners work in challenging and dangerous environments, and footwear that resists moisture, impacts, and prolonged wear is crucial for preventing injuries and ensuring worker safety. Proper safety footwear allows miners to perform their tasks with greater confidence and efficiency, as it minimises the risk of foot injuries from falling rocks and other hazards.

- **IS 1989 (Part 2) : 1986** — *Leather Safety Boots for Heavy Metal Industries*

In industries like steel and metalworking, where conditions involve high-impact hazards and exposure to extreme heat, safety boots must be designed to offer reliable protection, durability, and comfort. Workers in heavy metal industries face daily risks from heavy objects, high temperatures, and intense physical activity, making proper footwear essential to avoid injuries and enhance workplace safety.

- **IS 11226 : 1993** — *Leather Safety Boots with Moulded Rubber Sole*

Leather safety footwear with a Direct Moulded Rubber Sole (DMRS) is a popular choice for workers who need durable, comfortable, and protective footwear in demanding environments. These shoes combine the benefits of high-quality leather uppers with robust rubber soles, offering superior safety, comfort, and performance. The safety footwear covered in this standard may be used by the workers of mines, refineries, fertilizer plant or places where working surface is oily to protect workers from various occupational hazards.

- **IS 14544 : 2022** — *Leather Safety Boots with Moulded PVC Sole*

IS 14544:2022 is the Indian Standard that outlines the essential requirements for leather safety and protective footwear featuring direct moulded polymeric soles. This standard is crucial for footwear designed to safeguard users in various industrial settings, particularly where there is a risk of hazards such as impact, compression, and exposure to chemicals.

- **IS 5557 (Part 1) : 2024** — *All Rubber Gum Boots and Ankle Boots Part 1 Safety and Protective (Fifth Revision)*

This Indian Standard, IS 5557 (Part 1) : 2024, addresses the need for safety and protective all rubber gum boots and ankle boots used in various industrial and professional settings. These boots are essential for protecting the feet, ankle, and sometimes the knee, against rain, wet conditions, and slurry. They are also crucial in environments exposing wearers to chemicals, oil/grease, and in mining and explosion-oriented conditions.

- **IS 5557 (Part 2) : 201**8 — *All rubber gum boots and ankle boots: Part 2 occupational purposes*

Indian Standard IS 5557 (Part 2) covers All Rubber Gum Boots and Ankle Boots designed for occupational use. These boots protect your feet, ankles, and legs from water, mud, and chemicals, keeping you safe and comfortable in various environments like tanneries, food and beverage industries, sewage treatment plants, petrochemical industries, pharmaceutical industries, garbage disposal, municipal operations, cement and construction work, road building, horticulture, agriculture in cold areas, petrol pumps, and other oily areas.

- **IS 12254 : 2021** — *PVC Industrial Boots*

Polyvinyl chloride (PVC) boots are meant to protect feet, ankles and knees against rains during monsoon and while working in slushy area. Such boots are also recommended for light duty purpose in tanneries, food and beverage industries, sewage treatment plants, petrochemical, cement and pharmaceuticals industries, garbage disposal in municipal corporations and in building construction work, because polyvinyl chloride has very good resistance to water, alcohols, acids and alkalies.

GUIDE FOR SELECTION OF PERSONAL PROTECTIVE EQUIPMENT'S

Personal Protective Equipment (PPE) is critical for ensuring safety in environments where hazards exist. The selection of appropriate PPE depends on several factors, including the type of hazard, the level of exposure, and the specific tasks being performed. Below is a comprehensive guide to help select the right PPE
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List of Indian standard on Guide for Selection for Personal Protective Equipment's:

- **·S 10667 : 2025** — *Selection of Industrial Safety Equipment for Protection of Foot and Leg* — Guide The standard outlines different types of safety footwear and leg guards based on risk assessment. These include safety shoes with metal or composite toe caps, safety boots and rubber shoes for wet or slippery conditions, conductive and non-sparking shoes for hazardous environments, and foundry safety boots for protection against hot splashes.

- **IS 8519 : 2024** — *Guide for Selection of Occupational Protective Clothing* — Body Protection (Selection, Care and Maintenance) (First Revision) This Indian Standard, IS 8519: 2024, provides guidance on selecting, using, and caring for body protective clothing designed to shield individuals from chemical, physical, and biological hazards in the workplace.

- **IS 8807 : 2025** — *Selection of Industrial Safety Equipment for Protection of Arms and Hands* — Guidelines (First Revision) This Indian Standard provides a guide for selecting industrial safety equipment for the protection of arms and hands. It aims to assist workers and safety personnel in choosing appropriate protective gear against encountered hazards.

- **IS 8520 : 2023** — *Eye and Face Protection — Guidance on Selection, use, and Maintenance (First Revision)*

This Indian Standard, IS 8520 : 2023, offers guidance on the selection, use, and maintenance of eye and face protection. It aims to help specifiers and users control various hazards, including mechanical, chemical, optical radiation, and biological risks, in occupational and similar settings.

- **IS 9623 : 2008** — *Selection, Use and Maintenance of Respiratory Protective Devices — Code of Practice (First Revision)*

This Indian Standard provides a code of practice for the selection, use, and maintenance of respiratory protective devices (RPDs) in industries. It aims to guide employers and employees in ensuring adequate protection against inhaling airborne contaminants and oxygen-deficient atmospheres.